

LC 3660 : Jump Game IX

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Problem Statement & Rules

Basically, from any index **i** jump to any index **j**:

- **forward jump allowed** if going to smaller vol
- **backward jump allowed** if going to larger vol

need to return max possible value one can reach starting from every index

Example:

[2, 3, 1]

- for 2 we can go to 1 and then from 1 can go to 3
So [3, 3, 3]

Basic Idea

-> sabse pehle left to jaa hi sakte for larger value, say this was premax array

then

```
[2, 3, 1]
```

```
-> [2, 3, 3]
```

now we need to see if we can connect left part to right part

Walkthrough

```
nums = [2, 3, 1, 5, 4]
premax = [2, 3, 3, 5, 5]
```

at $i=1 \rightarrow$

- left side = [2, 3]
- curr = 3
- right side = [1, 5, 4]

```
suffixmin = 1
premax[curr] = 3
```

so $3 > 1$, one can jump from **3** $\rightarrow$ **1**

now 1 can also jump to a backward larger value

so **3** $\rightarrow$ **1** $\rightarrow$ **3**

hence left and right side are connected

once we are connected to the right side, we can access whatever max is reachable from the right side

hence $\text{ans}[i] = \text{ans}[i+1]$

thats why we also process from right to left.

C++ Solution

```
class Solution {
public:
    vector<int> maxValue(vector<int>& nums) {

        int n=nums.size();
        vector<int> ans(n);
        vector<int> premax(n,0);

        premax[0]=nums[0];
        for(int i=1;i<n;i++){
            premax[i]=max(premax[i-1],nums[i]);
        }

        int suffmin=INT_MAX;
        for(int i=n-1;i>=0;i--){
            if(premax[i]>suffmin){
                ans[i]=ans[i+1];
            }
            else{
                ans[i]=premax[i];
            }
            suffmin=min(suffmin,nums[i]);
        }
        return ans;
    }
};
```